Python Lab

1. Write a Python program to declare variables, perform arithmetic operations, and display results.

```python
# Take input from the user
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

# Perform arithmetic operations
addition = num1 + num2
subtraction = num1 - num2
multiplication = num1 * num2
division = num1 / num2
modulus = num1 % num2
exponent = num1 ** num2
floor_division = num1 // num2

# Display results
print("\n----- Results -----")
print("Number 1:", num1)
print("Number 2:", num2)
print("---------------------------")
print("Addition:", addition)
print("Subtraction:", subtraction)
print("Multiplication:", multiplication)
print("Division:", division)
print("Modulus:", modulus)
print("Exponent:", exponent)
print("Floor Division:", floor_division)
```

OUTPUT

2. Create a program to check if a number is even or odd using if-else.

```
# Take input from the user
num = int(input("Enter a number: "))

# Check if the number is even or odd
if num % 2 == 0:
    print(num, "is an Even number.")
else:
    print(num, "is an Odd number.")
```

OUTPUT

3. Write a Python program to print the first n Fibonacci numbers using a for loop.

```
# Take input from the user
n = int(input("Enter the number of terms: "))

# Initialize first two Fibonacci numbers
a, b = 0, 1

print("\nFibonacci Series:")

# Use a for loop to generate Fibonacci sequence
for i in range(n):
    print(a, end=" ")
    # Update values of a and b
    a, b = b, a + b
```

OUTPUT

4. Implement a program that accepts a string and counts the number of vowels and consonants.

```
# Take input from the user
text = input("Enter a string: ")

# Initialize counters
vowels = 0
consonants = 0

# Define vowel characters
vowel_chars = "aeiouAEIOU"

# Loop through each character in the string
for char in text:
    if char.isalpha():  # Check if the character is a letter
        if char in vowel_chars:
            vowels += 1
        else:
            consonants += 1

# Display the result
print("\n----- Result -----")
print("Input String:", text)
print("Number of vowels:", vowels)
print("Number of consonants:", consonants)
```

OUTPUT

5. Create a program to store student details in a dictionary and retrieve details based on user input.

```python
# Create an empty dictionary to store student details
students = {}

# Get the number of students
n = int(input("Enter number of students: "))

# Input details for each student
for i in range(n):
    print(f"\nEnter details for Student {i + 1}:")
    roll_no = input("Enter Roll Number: ")
    name = input("Enter Name: ")
    course = input("Enter Course: ")
    marks = float(input("Enter Marks: "))

    # Store details in dictionary
    students[roll_no] = {
        "Name": name,
        "Course": course,
        "Marks": marks
    }

# Retrieve details based on roll number
print("\n----- Retrieve Student Details -----")
search_roll = input("Enter Roll Number to search: ")

# Check if roll number exists
if search_roll in students:
    print("\nStudent Details Found:")
    print("Roll Number:", search_roll)
    print("Name:", students[search_roll]["Name"])
    print("Course:", students[search_roll]["Course"])
    print("Marks:", students[search_roll]["Marks"])
else:
    print("\nNo student found with Roll Number:", search_roll)
```

OUTPUT

6. Demonstrate the use of break, continue, and pass in loops.

```
print("=== Demonstration of 'break' statement ===")
for i in range(1, 6):
    if i == 4:
        print("Break encountered at i =", i)
        break  # exits the loop when i == 4
    print("Current value:", i)

print("\n=== Demonstration of 'continue' statement ===")
for i in range(1, 6):
    if i == 3:
        print("Continue encountered at i =", i)
        continue  # skips the rest of the code for i == 3
    print("Current value:", i)

print("\n=== Demonstration of 'pass' statement ===")
for i in range(1, 6):
    if i == 2:
        pass  # does nothing, acts as a placeholder
        print("Pass executed at i =", i)
    print("Current value:", i)
```

<u>OUTPUT</u>